

Hospital Appointment Database

```
CREATE DATABASE hospital_lab;  
USE hospital_lab;
```

Patients

```
CREATE TABLE patients (  
    patient_id INT PRIMARY KEY,  
    patient_name VARCHAR(100),  
    age INT,  
    city VARCHAR(50)  
);  
  
INSERT INTO patients VALUES  
(1, 'Rohan Das', 34, 'Hyderabad'),  
(2, 'Meena Iyer', 46, 'Chennai'),  
(3, 'Kabir Khan', 29, 'Hyderabad'),  
(4, 'Lakshmi Rao', 61, 'Bangalore'),  
(5, 'John Mathew', 38, 'Mumbai'),  
(6, 'Ayesha Ali', 25, NULL),  
(7, 'Naveen Reddy', 52, 'Pune');
```

Doctors

```
CREATE TABLE doctors (  
    doctor_id INT PRIMARY KEY,  
    doctor_name VARCHAR(100),  
    specialization VARCHAR(50),  
    consultation_fee DECIMAL(10,2)  
);  
  
INSERT INTO doctors VALUES  
(101, 'Dr. Sharma', 'Cardiology', 1200),  
(102, 'Dr. Farah', 'Dermatology', 800),  
(103, 'Dr. Joseph', 'Orthopedics', 1000),  
(104, 'Dr. Mehta', 'General Medicine', 600),
```

```
(105, 'Dr. Sana', 'Neurology', 1500),  
(106, 'Dr. Rao', 'Pediatrics', 700);
```

Appointments

```
CREATE TABLE appointments (  
    appointment_id INT PRIMARY KEY,  
    patient_id INT,  
    doctor_id INT,  
    appointment_date DATE,  
    status VARCHAR(30)  
);  
  
INSERT INTO appointments VALUES  
(1001, 1, 101, '2026-10-01', 'Completed'),  
(1002, 2, 104, '2026-10-01', 'Completed'),  
(1003, 3, 102, '2026-10-02', 'Cancelled'),  
(1004, 1, 105, '2026-10-03', 'Completed'),  
(1005, 4, 101, '2026-10-03', 'Scheduled'),  
(1006, 5, 103, '2026-10-04', 'Completed'),  
(1007, 3, 104, '2026-10-04', 'Completed'),  
(1008, NULL, 102, '2026-10-05', 'Scheduled'),  
(1009, 20, 103, '2026-10-05', 'Completed'),  
(1010, 2, NULL, '2026-10-06', 'Scheduled');
```

This deliberately gives you:

- patients with no appointments,
- doctors with no appointments,
- an appointment with no patient,
- an appointment with an invalid patient ID,
- an appointment with no doctor,
- actual `NULL` data.

Exercises

1. Display appointment ID, patient name, doctor name and appointment date.
2. Show the doctor and specialization for every completed appointment.

3. Find all appointments made by patients from Hyderabad.
4. Display every patient and their appointment details. Patients who have never booked an appointment must also appear.
5. Find patients who have **never booked an appointment**.
6. Display every doctor and appointments handled by them, including doctors who have never handled an appointment.
7. Find doctors who currently have **no appointments**.
8. Identify appointment records for which a valid patient cannot be found.
9. Identify appointment records for which a doctor has not been assigned.
10. Display:

Patient Name
Doctor Name
Specialization
Consultation Fee
Appointment Status

for all valid appointments.

11. Find the number of appointments handled by each doctor.
12. Find the total consultation value generated by each doctor based on **completed appointments only**.
13. Find doctors who have handled more than one appointment.
14. Find the specialization generating the highest total consultation value.
15. Display each patient and the number of appointments they have booked, including patients with zero appointments.
16. Find patients who have consulted more than one doctor.
17. Find patients who have visited a **Cardiology** doctor.
18. Display appointments where the consultation fee is greater than ₹900.
19. Find the average consultation fee of doctors involved in completed appointments.

20. Display the doctor who has handled the highest number of completed appointments.